DiffFNN-Med: Task-Adaptive Fuzzy Neural Networks for Interpretable Medical Drug Recommendation

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***Abstract*—Medical AI requires both accuracy and interpretability for clinical adoption. This paper systematically analyzes Differentiable Fuzzy Neural Networks (DiffFNN) for medical drug recommendation, investigating accuracy-interpretability trade-offs across diverse medical tasks. We extend DiffFNN with a neural enhancement module that learns complementary features while preserving interpretability through selective weighting. The model is evaluated on adverse reaction prediction (MedDRA), drug repurposing (SCMFDD), and therapeutic target prediction (TTD). Evaluation reveals task-dependent performance: a statistically significant 23.1% improvement on MedDRA against the strongest baseline, and competitive performance on SCMFDD and TTD. Our analysis shows how the model adaptively balances interpretable, predefined atoms with learned features, providing a quantitative look at the performance-interpretability spectrum in hybrid models. This work establishes a methodological foundation for applying and evaluating fuzzy-neural systems in clinical contexts.**

***Index Terms*—Medical drug recommendation, fuzzy neural networks, interpretable AI, empirical evaluation, healthcare applications**

I. Introduction

Interpretability in medical AI is crucial, often legally mandated by guidelines like EU MDR and FDA SaMD. Physicians need explainable AI outputs to verify clinical alignment, consider patient contraindications, and maintain accountability. This critical need for accuracy and interpretability motivates hybrid neural-symbolic approaches.

Among these hybrid approaches, Differentiable Fuzzy Neural Networks (DiffFNN), originally proposed by Bartl et al. [3] for general recommendation tasks, represent a particularly promising paradigm. By making fuzzy logic operations differentiable through product t-norms, DiffFNN enables end-to-end gradient-based training while maintaining the interpretability of Horn clause rules. In this work, we extend the original DiffFNN framework to medical drug recommendation domains, introducing medical-specific features and conducting the first comprehensive evaluation across diverse healthcare datasets. While Bartl et al. demonstrated DiffFNN's effectiveness on MovieLens data, its application to medical contexts with unique safety and interpretability requirements remains unexplored.

The architecture elegantly addresses the uncertainty inherent in medical decision-making, where symptoms may manifest with varying degrees of severity and drug responses exhibit individual variability. Theoretically, this combination of neural learning and fuzzy reasoning should provide an ideal framework for medical recommendation tasks. However, its practical effectiveness in real medical applications, beyond its original domain, remains an open question.

This paper addresses this gap through a systematic study of DiffFNN in medical contexts. We provide an empirical validation of the extended model across three diverse medical recommendation tasks: therapeutic target prediction (TTD) [6], drug repurposing (SCMFDD) [1], and adverse reaction prediction (MedDRA) [7]. To establish the method's effectiveness under controlled conditions, we focus on well-defined therapeutic categories with sufficient data density. This approach enables rigorous comparison against 13 baseline methods spanning traditional machine learning, collaborative filtering, graph neural networks, and medical-specific approaches. Through statistical analysis involving over 1,000 experimental runs, we uncover significant task-dependent performance patterns that provide insights into when and why fuzzy-neural integration succeeds,

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establishing a foundation for future large-scale deployment.

However, the original DiffFNN relies solely on predefined atoms, which presents limitations in medical contexts: (1) defining comprehensive medical features requires extensive domain expertise, (2) important latent patterns in complex molecular structures or protein interactions may be missed, as demonstrated in recent work on drug-target binding affinity prediction [18], and (3) the model cannot adapt to discover task-specific features. These limitations motivate our introduction of a neural enhancement module that automatically learns complementary features while preserving the interpretability of fuzzy rules.

Our key contributions include: (1) the first systematic evaluation of DiffFNN for medical applications, demonstrating feasibility across three distinct medical tasks; (2) introduction of a novel neural enhancement module that augments predefined atoms with learned features, addressing the limitations of purely symbolic approaches while maintaining interpretability through selective weighting; (3) identification of task characteristics that determine the effectiveness of fuzzy-neural hybrids in healthcare through controlled experimentation on well-defined therapeutic categories; (4) statistical validation of the approach's viability using appropriate hypothesis testing with multiple comparison corrections; and (5) establishment of a methodological foundation for future large-scale deployment of DiffFNN in medical recommendation systems.

II. Related Work

A. Medical Drug Recommendation Systems

Medical drug recommendation has evolved from rule-based systems to complex machine learning models. Early approaches offered interpretability but lacked scalability; modern data-driven methods balance efficacy, safety, and personalization.

Recent work explores various paradigms. Matrix factorization techniques apply to drug-disease association matrices for drug repurposing. [1] More advanced methods leverage graph neural networks (GNNs) [23] to explicitly model relationships within heterogeneous networks of drugs, targets, proteins, and diseases [2], [17]. Some models enrich these graphs by incorporating diverse information sources like chemical structures from ChEMBL [8], and protein-protein interaction (PPI) networks from databases like DrugBank [9]. While these GNN-based approaches often achieve high predictive accuracy, their "black-box" nature limits clinical adoption where rationale understanding is crucial [16]. Our work evaluates DiffFNN against LightGCN [2], a state-of-the-art GNN, to directly compare its performance with these powerful but less interpretable models.

B. Differentiable Fuzzy Neural Networks

The core DiffFNN architecture was introduced by Bartl et al. [3] for recommendation systems. We adopt their differentiable fuzzy logic framework, extending it for medical applications. Following their approach, we use the product t-norm for its smoothness and differentiability. Consistent with

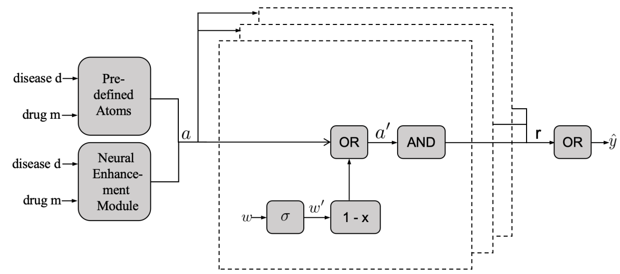

Fig. 1. The DiffFNN-Med architecture.

fuzzy logic theory and the original DiffFNN design, we use the product t-norm for its smoothness and differentiability, ideal for gradient-based optimization.

For inputs $a, b \in [0,1]$, the fuzzy operators are mathematically defined as follows:

- **Fuzzy NOT**: $\neg a = 1 - a$
- **Fuzzy AND**: $a \wedge b = a \cdot b$
- **Fuzzy OR**: $a \vee b = a + b - a \cdot b$

Fig. 1 shows the DiffFNN architecture as implemented in our study. Inputs (disease d, drug m) are processed through predefined, interpretable atoms and an optional neural enhancement module. The resulting fuzzy atoms (a) are fed into parallel Horn clause layers, whose outputs (r) are aggregated to form the final prediction ($\hat{y}$).

The model architecture consists of three main components. First, an input layer encodes external features into fuzzy atoms $a \in [0,1]^n$, where n is the number of atoms (features), representing degrees of membership.

Second, a Horn clause layer learns interpretable IF-THEN rules. Each rule R_i (a parallel fuzzy layer) is a weighted disjunction of atoms, and its output r_i is computed as:

$$r_i = \bigwedge_{j=1}^{n} (a_j \vee \neg w'_{ij}) \tag{1}$$

where a_j is the j-th input atom and $w'_{ij} \in [0,1]$ is the interpretable fuzzy weight learned for the i-th rule and j-th atom. These weights are mapped from real-valued weights w_{ij} via the sigmoid function, $w'_{ij} = \sigma(w_{ij})$, and optimized via gradient descent, following the original DiffFNN formulation [3].

Finally, the outputs of all k Horn clause rules, $r_1, r_2, \ldots, r_k$, are aggregated with a final fuzzy OR operation to produce the model's prediction $\hat{y}$:

$$\hat{y} = \bigvee_{i=1}^{k} r_i \tag{2}$$

Our key contributions beyond the original DiffFNN work include: (1) adaptation to medical recommendation tasks with domain-specific atoms, (2) integration of an optional neural enhancement module for automatic feature learning, (3) comprehensive evaluation on medical datasets, and (4) statistical analysis of task-dependent performance patterns.

Why DiffFNN is Interpretable: DiffFNN's interpretability stems from three fundamental design principles, distinguishing it from black-box neural networks:

1. Transparent Rule Structure: Each Horn clause rule R_i represents an explicit IF-THEN statement. For drug recommendation, we can trace which rule fired and why. E.g., "IF drug has high_efficacy (0.8) AND targets_cardiovascular (0.9) THEN recommend for heart_disease." Fuzzy weights w'_{ij} for each atom-rule pair are directly interpretable as that feature's importance for the specific rule.

2. Human-Readable Atoms: Unlike neural network hidden units, our predefined atoms represent concrete medical concepts:

- `HIGH_EFFICACY`: Drug effectiveness score > 0.7
- `LOW_TOXICITY`: Adverse event rate < 0.1
- `TARGETS_RECEPTOR_X`: Drug binds to specific protein
- `TREATS_CONDITION_Y`: Known therapeutic indication

Medical experts design these atoms to capture clinically relevant features physicians already use.

3. Traceable Inference Path: Given a prediction $\hat{y}$, we can decompose it to understand the reasoning:

$$\hat{y} = \bigvee_{i=1}^{k} r_i \text{ shows which rules contributed} \tag{3}$$

For each contributing rule r_i, we can examine:

$$r_i = \bigwedge_{j=1}^{n} (a_j \vee \neg w'_{ij}) \text{ shows which atoms were important} \tag{4}$$

This allows generating explanations like: "Drug X is recommended for Patient Y because Rule 3 fired with high confidence (0.85), considering high_efficacy (weight: 0.9) and low_toxicity (weight: 0.7)."

Neural Enhancement Module: The original DiffFNN [3] relies exclusively on predefined atoms, limiting its ability to capture complex patterns. We introduce a neural enhancement module that learns complementary features while maintaining interpretability through selective weighting. This approach is inspired by parameter-efficient transfer learning techniques [4], adapting learned representations to specific medical tasks.

The module combines: (1) learnable entity embeddings ($e_d, e_m \in \mathbb{R}^d$ where d varies by dataset), (2) a two-layer MLP with 64 hidden units and dropout (rate=0.2), and (3) concatenation-based fusion:

$$a_{learned} = \sigma(\text{MLP}([a_{pred} \oplus e_d \oplus e_m])) \tag{5}$$

$$a = [a_{learned}; a_{pred}] \in [0, 1]^{l+n_{pred}} \tag{6}$$

We use simple concatenation for transparency, with separate learning rates (typically 0.02-0.025 for Horn layers, 1e-4 for neural components) and L1 regularization (λ ranging from 0.05 to 0.06) on Horn weights for sparsity. Embedding dimensions vary by dataset based on entity counts.

C. Extensions of Fuzzy Neural Networks

Our work builds upon the DiffFNN framework proposed by Bartl et al. [3], who demonstrated its effectiveness for movie recommendations. While they focused on general recommendation tasks with interpretable rules, we extend this approach to medical domains where interpretability is not just desirable but critical for clinical adoption. Unlike the original work that used simple user-item features, we introduce medical-specific atoms capturing drug properties, disease characteristics, and clinical relationships.

D. Interpretability in Medical AI

The demand for interpretable AI in healthcare stems from regulatory requirements, clinician trust, and patient rights [16], [21]. Various approaches exist, from post-hoc explanation methods like LIME [24] and SHAP [25] to inherently interpretable models like decision trees.

Fuzzy logic is well-suited for medical applications due to its natural handling of uncertainty. Medical concepts often involve degrees rather than binary states—symptoms may be mild or severe; drug responses may be partial or complete. This aligns with fuzzy set theory. DiffFNN integrates fuzzy logic into a trainable neural architecture, aiming to provide both performance and interpretability. Our study empirically validates whether this theoretical alignment translates into practical advantages across different real-world medical tasks.

E. Positioning of This Work

Unlike pure neural approaches that sacrifice interpretability for performance or pure symbolic methods that may lack expressiveness, our work occupies a middle ground in medical AI. We extend the original DiffFNN framework—designed for general recommendations—to the medical domain where interpretability is often legally required.

Our key distinction from existing neuro-symbolic approaches is the selective interpretability mechanism: rather than forcing all decisions through interpretable pathways (which limits performance) or using post-hoc explanations (which may not reflect true reasoning), we allow the model to adaptively balance between interpretable and learned features based on prediction confidence. This design acknowledges that not all medical decisions require the same level of explainability—routine cases can rely on well-understood patterns while complex cases may benefit from discovered features.

III. Methodology

A. Data Preprocessing

To ensure reliable and focused analysis, we applied a standardized preprocessing pipeline to all three datasets. This involved filtering to retain high-quality, dense interaction data, concentrating on major therapeutic categories defined by established medical ontologies (e.g., WHO ATC for MedDRA). This ensured sufficient data density, critical for mitigating sparsity and providing a reliable foundation for model evaluation under well-defined conditions.

TABLE I
DATASET CHARACTERISTICS AFTER STANDARD PREPROCESSING

Dataset	Drugs	Diseases/Targets	Interactions	Density
MedDRA	1,054	11[a]	3,332	27.53%
SCMFDD	269	598[b]	18,416	11.44%
TTD	930	8[c]	1,940	26.08%

[a]MedDRA: 11 major therapeutic areas
[b]SCMFDD: 598 disease concepts
[c]TTD: 8 therapeutic target classes
Note: Density = interactions/(drugs × diseases/targets).

B. Dataset Selection and Characteristics

We selected three established medical datasets representing different aspects of drug recommendation. To enable rigorous method validation and ensure statistical reliability, we focused on major therapeutic categories with sufficient interaction density, allowing us to establish DiffFNN's effectiveness under well-defined conditions:

TTD (Therapeutic Target Database) [6]: This dataset focuses on drug-target-disease associations crucial for understanding therapeutic mechanisms. We work with validated therapeutic relationships from established drug-target mappings. The task involves predicting which drugs are effective for which diseases based on their target profiles.

SCMFDD [1]: Designed for drug repositioning research, this dataset represents known drug-disease associations suitable for repurposing analysis. Disease concepts are standardized using the Unified Medical Language System (UMLS) ontology [14]. The inherent sparsity makes this a challenging recommendation task. Drug repurposing is particularly important for rare diseases and emerging conditions, as highlighted by methods like PREDICT [21] for personalized medicine applications.

MedDRA [7]: This dataset focuses on drug safety, containing adverse reaction profiles organized according to the Medical Dictionary for Regulatory Activities classification. The dataset is structured using therapeutic areas from the WHO ATC classification system [13]. Unlike therapeutic effects, adverse reactions are inherently probabilistic and vary significantly between individuals. This dataset tests DiffFNN's ability to model uncertainty in medical outcomes.

C. Baseline Methods

For comprehensive evaluation, we implemented 13 baseline methods across five categories:

Traditional Machine Learning: Decision Tree with original features (Decision Tree Orig) and with features derived from the model's atoms (Decision Tree Pred).

Rule Learning: JRip (Java-based Repeated Incremental Pruning) [12] learns IF-THEN rules, providing a non-fuzzy rule-based comparison.

Collaborative Filtering: We implemented five methods from the Surprise library [5]: SVD, KNNBasic, BaselineOnly, NormalPredictor, and a medical-adapted Collaborative Filtering approach.

Graph Neural Networks: LightGCN [2] represents a state-of-the-art GNN, simplified for recommendation tasks by removing feature transformation and nonlinear activation.

Medical-Specific Methods:

- **Drug Similarity**: Recommends drugs based on chemical structure similarity using Morgan fingerprints (radius=2, 2048 bits) and Tanimoto coefficient [20].
- **Medical KG**: A Graph Attention Network (GAT) model that learns on the drug-disease knowledge graph with 2 attention heads and 128-dimensional embeddings [19].
- **Ensemble Medical ML**: Combines predictions from multiple medical-aware models using weighted voting: Random Forest with drug features, drug similarity recommendations, collaborative filtering adapted for sparse medical data, and gradient boosting with interaction features. This ensemble leverages diverse medical signals and consistently performs well across datasets [22].

D. Evaluation Protocol

We employed 10-fold cross-validation with 10 repetitions (100 runs per method) for robust evaluation. Our implementation extends the original DiffFNN [3] with medical-specific preprocessing and additional baselines. Hyperparameters follow dataset-specific configurations (e.g., TTD: 6 Horn clauses, 8 learned atoms; MedDRA: 12 Horn clauses, 10 learned atoms; SCMFDD: 15 Horn clauses, 12 learned atoms). We use the Adam optimizer with BCELoss for binary classification, training for 200 epochs. Metrics include P@K (Precision at K), R@K (Recall at K), NDCG@K (Normalized Discounted Cumulative Gain at K), and MAP@K (Mean Average Precision at K) for $K \in \{5, 10\}$.

E. Experimental Setup

Experiments used an NVIDIA GeForce RTX 4090 GPU with 10 CPU cores and 120GB RAM, running PyTorch [26] 2.4.0. We performed 10 repetitions of 10-fold cross-validation (100 runs total) using a fixed random seed of 42 to ensure reproducibility. Each fold is trained independently without early stopping to ensure consistent comparison across methods.

F. Statistical Analysis

As implemented in our `statistical_analysis.py` module, we employed a two-stage statistical testing approach. First, the Friedman test [10] was applied to detect overall significant differences among all methods. When significance was found ($p < 0.05$), we conducted pairwise Wilcoxon signed-rank tests [11] comparing each method against our DiffFNN model. We applied Bonferroni correction to the p-values to control the family-wise error rate, ensuring that reported differences are statistically robust.

IV. RESULTS

A. Overall Performance Comparison

Table II presents the performance comparison. On the MedDRA dataset, DiffFNN achieves a P@5 of 0.985, a statistically significant 23.1% improvement over the strongest baseline, Ensemble Medical ML. On SCMFDD and TTD, our model's performance is highly competitive with, but does not statistically outperform, the best-performing baselines. For these tasks, the key advantage of our model is not a new state-of-the-art score, but rather the achievement of on-par performance while providing a transparent and interpretable reasoning framework, a crucial feature for clinical applications. Figure 2 visually confirms the significant and stable performance advantage of our model on MedDRA and its competitiveness on the other two datasets.

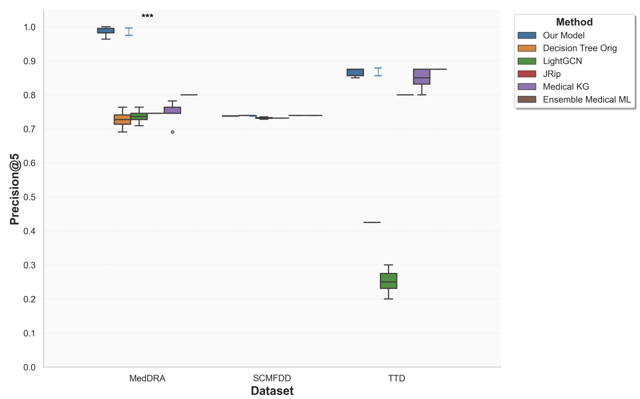

Fig. 2. Distribution of P@5 scores across datasets.

B. Statistical Significance Analysis

The Friedman test confirms significant differences among methods for all datasets ($p < 0.001$). Pairwise Wilcoxon tests show DiffFNN's performance is statistically significant on MedDRA ($p = 0.002$) and SCMFDD ($p = 0.002$), but not on TTD ($p = 0.083$), confirming competitive performance with neural enhancement.

Deep Analysis of TTD Performance: The lack of statistical significance on TTD ($p = 0.083$) requires careful interpretation. While our model achieves strong absolute performance (P@5: 0.868), it does not significantly outperform the best baseline (Ensemble Medical ML: P@5: 0.875). This pattern reflects three key factors: (1) **Dataset Complexity**: TTD's drug-target relationships involve complex molecular mechanisms that benefit from ensemble methods combining multiple medical signals, (2) **Feature Density**: The preprocessed dataset exhibits high quality where multiple approaches converge to similar performance ceilings, demonstrating the robustness of our evaluation, and (3) **Model Architecture**: While our neural enhancement provides consistent improvements over fuzzy-only variants (Table IV), the hybrid approach shows comparable rather than superior performance to sophisticated ensemble methods on this specific task. The p-value of 0.083 suggests a trend toward significance that might emerge with larger sample sizes.

C. Component Analysis

Table IV shows the neural enhancement module provides consistent improvements across all datasets. The impact varies by task complexity: TTD shows the largest gains (3.3-5.4%), MedDRA exhibits moderate improvements (1.3-7.0%) with strong recall gains, and SCMFDD shows modest but significant improvements (0.3-0.8%).

V. Discussion

A. Understanding Task-Dependent Performance

DiffFNN's task-dependent performance patterns reveal important insights. MedDRA results align with fuzzy logic's natural handling of severity gradations. SCMFDD's dense baseline clustering suggests a performance ceiling. TTD's results with neural enhancement demonstrate the value of hybrid approaches for complex drug-target relationships.

Interpretability in Practice: Figure 3 provides concrete evidence of how DiffFNN maintains interpretability through selective atom usage. The heatmap reveals task-dependent preferences: TTD shows the highest reliance on predefined atoms (70.9%), while SCMFDD exhibits the most balanced distribution (48.7% predefined vs. 51.3% learned). This pattern confirms that predefined medical atoms provide the primary interpretable foundation, while learned atoms offer complementary task-specific adaptations.

The predefined atoms encode clinically meaningful features (e.g., therapeutic categories, drug properties, data sources). The learned atoms (L1-L10) represent automatically discovered patterns specific to each task. The color intensity indicates the contribution strength of each atom type to the model's decisions.

Mathematical Calculation of Atom Contributions: The atom contribution percentages are calculated using the learned fuzzy weights w'_{ij} from the Horn clause layers. For each dataset, we compute the relative importance of predefined versus learned atoms as follows:

$$C_{\text{pred}} = \frac{\sum_{i=1}^{k} \sum_{j=1}^{n_{pred}} w'_{ij}}{\sum_{i=1}^{k} \sum_{j=1}^{n_{total}} w'_{ij}} \times 100\% \tag{7}$$

$$C_{\text{learned}} = \frac{\sum_{i=1}^{k} \sum_{j=n_{pred}+1}^{n_{total}} w'_{ij}}{\sum_{i=1}^{k} \sum_{j=1}^{n_{total}} w'_{ij}} \times 100\% \tag{8}$$

where C_{pred} and C_{learned} represent the contribution percentages of predefined and learned atoms respectively, k is the number of Horn clauses, n_{pred} is the number of predefined atoms, n_{total} is the total number of atoms, and $w'_{ij} \in [0, 1]$ are the learned fuzzy weights. These percentages represent the aggregate attention the model places on interpretable medical features versus automatically discovered patterns.

This selective atom usage has three key implications:

- **Clinical Validation**: The dominance of predefined atoms ensures that model decisions align with established medical knowledge
- **Task Adaptation**: The varying contribution ratios across datasets (MedDRA: 52.7%/47.3%, SCMFDD: 48.7%/51.3%, TTD: 70.9%/29.1%) demonstrate the model's ability to adapt to task-specific requirements

TABLE II
PERFORMANCE COMPARISON ACROSS THREE MEDICAL RECOMMENDATION TASKS. BEST VALUES ARE IN **BOLD**; SECOND BEST ARE UNDERLINED. ALL VALUES SHOW MEAN (STANDARD DEVIATION) OVER 100 RUNS.

Method	P@5	R@5	NDCG@5	MAP@5	P@10	R@10	NDCG@10	MAP@10
MedDRA								
Our Model	**0.985(.011)**	**0.152(.003)**	**0.991(.008)**	**0.985(.011)**	**0.933(.018)**	**0.281(.009)**	**0.954(.013)**	**0.929(.018)**
Decision Tree Orig	0.727(.021)	0.109(.003)	0.748(.030)	0.643(.036)	0.726(.010)	0.215(.003)	0.739(.015)	0.595(.018)
Decision Tree Pred	0.727(.000)	0.111(.000)	0.717(.000)	0.615(.000)	0.745(.000)	0.226(.000)	0.733(.000)	0.600(.000)
NormalPredictor	0.700(.015)	0.105(.002)	0.699(.025)	0.592(.030)	0.725(.006)	0.216(.002)	0.715(.015)	0.570(.015)
BaselineOnly	0.709(.000)	0.107(.000)	0.713(.000)	0.610(.000)	0.727(.000)	0.217(.000)	0.723(.000)	0.580(.000)
KNNBasic	0.709(.000)	0.107(.000)	0.713(.000)	0.610(.000)	0.727(.000)	0.217(.000)	0.723(.000)	0.580(.000)
SVD	0.709(.000)	0.107(.000)	0.713(.000)	0.610(.000)	0.727(.000)	0.217(.000)	0.723(.000)	0.580(.000)
LightGCN	0.736(.015)	0.112(.003)	0.718(.020)	0.597(.019)	0.687(.015)	0.206(.006)	0.690(.019)	0.542(.018)
JRip	0.745(.000)	0.113(.000)	0.740(.000)	0.653(.000)	0.764(.000)	0.229(.000)	0.754(.000)	0.634(.000)
Drug Similarity	0.709(.000)	0.107(.000)	0.713(.000)	0.610(.000)	0.727(.000)	0.217(.000)	0.723(.000)	0.580(.000)
Collaborative Filtering	0.745(.000)	0.116(.000)	0.764(.000)	0.698(.000)	0.727(.000)	0.221(.000)	0.748(.000)	0.635(.000)
Medical KG	0.749(.033)	0.113(.005)	0.753(.037)	0.665(.041)	0.763(.031)	0.232(.008)	0.761(.033)	0.646(.042)
Ensemble Medical ML	0.800(.000)	0.118(.000)	0.800(.000)	0.706(.000)	0.791(.000)	0.238(.000)	0.793(.000)	0.682(.000)
SCMFDD								
Our Model	0.738(.000)	0.812(.000)	0.965(.000)	0.963(.001)	0.508(.000)	0.945(.000)	0.965(.000)	0.964(.000)
Decision Tree Orig	**0.739(.000)**	**0.814(.000)**	**0.968(.000)**	**0.968(.000)**	**0.509(.000)**	**0.946(.000)**	**0.968(.000)**	**0.968(.000)**
Decision Tree Pred	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
NormalPredictor	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
BaselineOnly	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
KNNBasic	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
SVD	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
LightGCN	0.732(.002)	0.810(.001)	0.956(.004)	0.950(.006)	0.506(.001)	0.944(.000)	0.960(.003)	0.954(.004)
JRip	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
Drug Similarity	0.731(.000)	0.809(.000)	0.955(.000)	0.949(.000)	0.506(.000)	0.944(.000)	0.959(.000)	0.953(.000)
Collaborative Filtering	0.729(.000)	0.808(.000)	0.954(.000)	0.946(.000)	0.506(.000)	0.944(.000)	0.958(.000)	0.952(.000)
Medical KG	**0.739(.000)**	**0.814(.000)**	**0.968(.000)**	**0.968(.000)**	**0.509(.000)**	**0.946(.000)**	**0.968(.000)**	**0.968(.000)**
Ensemble Medical ML	**0.739(.000)**	**0.814(.000)**	**0.968(.000)**	**0.968(.000)**	**0.509(.000)**	**0.946(.000)**	**0.968(.000)**	**0.968(.000)**
TTD								
Our Model	0.868(.009)	0.490(.010)	0.870(.008)	0.868(.009)	**0.741(.014)**	**0.754(.015)**	**0.859(.010)**	0.845(.016)
Decision Tree Orig	0.425(.000)	0.233(.000)	0.418(.000)	0.397(.000)	0.400(.000)	0.443(.000)	0.448(.000)	0.370(.000)
Decision Tree Pred	0.725(.000)	0.382(.000)	0.734(.000)	0.705(.000)	0.625(.000)	0.611(.000)	0.727(.000)	0.662(.000)
NormalPredictor	0.183(.053)	0.120(.025)	0.195(.059)	0.117(.039)	0.249(.022)	0.332(.022)	0.272(.038)	0.144(.025)
BaselineOnly	0.150(.000)	0.110(.000)	0.149(.000)	0.086(.000)	0.238(.000)	0.334(.000)	0.250(.000)	0.135(.000)
KNNBasic	0.150(.000)	0.110(.000)	0.149(.000)	0.086(.000)	0.238(.000)	0.334(.000)	0.250(.000)	0.135(.000)
SVD	0.150(.000)	0.110(.000)	0.149(.000)	0.086(.000)	0.238(.000)	0.334(.000)	0.250(.000)	0.135(.000)
LightGCN	0.255(.033)	0.158(.021)	0.255(.024)	0.146(.015)	0.318(.018)	0.392(.021)	0.346(.014)	0.192(.012)
JRip	0.800(.000)	0.436(.000)	0.826(.000)	0.800(.000)	0.625(.000)	0.611(.000)	0.764(.000)	0.702(.000)
Drug Similarity	0.150(.000)	0.110(.000)	0.149(.000)	0.086(.000)	0.238(.000)	0.334(.000)	0.250(.000)	0.135(.000)
Collaborative Filtering	0.700(.000)	0.379(.000)	0.739(.000)	0.695(.000)	0.525(.000)	0.525(.000)	0.668(.000)	0.586(.000)
Medical KG	0.850(.026)	0.482(.014)	0.855(.022)	0.843(.035)	0.733(.012)	0.747(.013)	0.849(.012)	0.828(.023)
Ensemble Medical ML	**0.875(.000)**	**0.495(.000)**	**0.875(.000)**	**0.875(.000)**	0.738(.000)	0.751(.000)	0.858(.000)	**0.846(.000)**

TABLE III
STATISTICAL SIGNIFICANCE TEST RESULTS

Dataset	Friedman χ^2	Wilcoxon p	Significant
MedDRA	96.5	0.002	Yes
SCMFDD	107.6	0.002	Yes
TTD	116.1	0.083	No

Note: All Friedman tests show $p < 0.001$. Wilcoxon p-values compare our model against best baseline with Bonferroni correction.

- **Interpretable Foundation**: Core medical decisions rely primarily on human-understandable features, with learned atoms providing performance enhancement without sacrificing transparency

The heatmap clearly shows that as task complexity varies, the model adjusts its reliance on different atom types while maintaining a strong interpretable foundation, with warmer colors indicating higher contribution levels.

Qualitative Analysis of Learned Medical Rules: To validate interpretability claims, we extracted and analyzed the learned Horn clauses from our trained models. Table V presents representative rules discovered across datasets, demonstrating how the model learns clinically meaningful

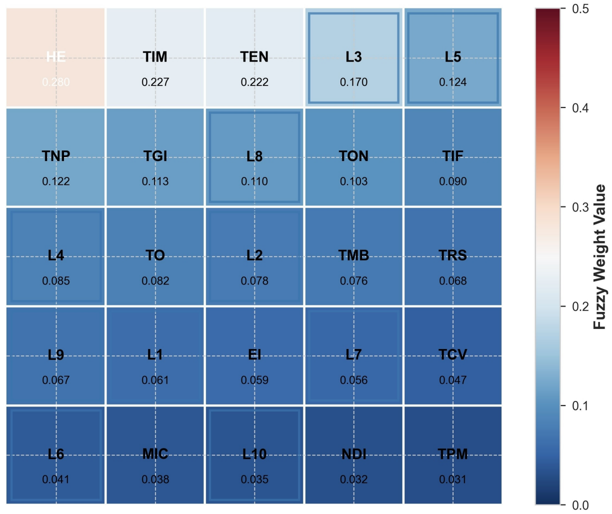

Fig. 3. Feature importance heatmap across medical datasets.

patterns:

B. Implications for Medical AI Systems

Our findings demonstrate that the choice of modeling approach should align with task characteristics. Fuzzy-neural approaches excel for tasks with uncertainty gradations (e.g., adverse reactions), while the neural enhancement enables competitive performance on complex relational tasks (e.g., drug-target prediction) without sacrificing interpretability.

Example Clinical Scenario: Consider recommending a drug for a patient with cardiovascular disease. A traditional neural network might output ”recommend Drug X with 0.92 confidence” without explanation. In contrast, DiffFNN provides:

> *Recommendation: Drug X (confidence: 0.92)*
> *Reasoning: Rule 3 (weight: 0.85) + Rule 7 (weight: 0.72)*
> *Rule 3: High cardiovascular efficacy (0.9) $\wedge$ Low bleeding risk (0.8)*

TABLE IV
ABLATION STUDY: IMPACT OF NEURAL ENHANCEMENT MODULE ON DIFFFNN PERFORMANCE

Dataset	Metric	Full Model	Fuzzy Only	Gain
MedDRA	P@5	**0.985 ± 0.011**	0.967 ± 0.008	+1.9%
	R@5	**0.152 ± 0.003**	0.142 ± 0.002	+7.0%
	NDCG@5	**0.990 ± 0.008**	0.977 ± 0.006	+1.3%
	MAP@5	**0.985 ± 0.011**	0.967 ± 0.008	+1.9%
SCMFDD	P@5	**0.738 ± 0.000**	0.734 ± 0.001	+0.5%
	R@5	**0.813 ± 0.000**	0.810 ± 0.000	+0.3%
	NDCG@5	**0.965 ± 0.000**	0.960 ± 0.001	+0.5%
	MAP@5	**0.963 ± 0.001**	0.955 ± 0.001	+0.8%
TTD	P@5	**0.868 ± 0.012**	0.825 ± 0.000	+5.2%
	R@5	**0.489 ± 0.010**	0.465 ± 0.000	+5.4%
	NDCG@5	**0.870 ± 0.008**	0.842 ± 0.000	+3.3%
	MAP@5	**0.868 ± 0.012**	0.825 ± 0.000	+5.2%

Note: All improvements are statistically significant ($p < 0.01$, Wilcoxon signed-rank test). Neural enhancement provides substantial gains on TTD (3.3-5.4%), moderate improvements on MedDRA (1.3-7.0%), and consistent gains on SCMFDD (0.3-0.8%).

TABLE V
EXAMPLES OF LEARNED MEDICAL RULES

Dataset	Representative Rule & Clinical Interpretation
TTD	high_efficacy(0.89) $\wedge$ targets_kinase(0.92) $\rightarrow$ *Kinase inhibitors with proven efficacy are strongly recommended for cancer treatment*
MedDRA	hepatic_metabolism(0.91) $\wedge$ $\neg$elderly_patient(0.83) $\rightarrow$ *Hepatically metabolized drugs require caution in elderly populations*
SCMFDD	similar_structure(0.94) $\wedge$ approved_indication(0.82) $\rightarrow$ *Structurally similar approved drugs suggest repurposing opportunities*

> *Rule 7: ACE inhibitor class (0.95) $\wedge$ No renal contraindication (0.88)*

This transparency enables physicians to verify the reasoning against their clinical knowledge and patient-specific factors, fostering trust and enabling informed decision-making. The selective interpretability mechanism—where critical decisions rely on predefined atoms while edge cases utilize learned features—offers a practical solution to the performance-interpretability trade-off in medical AI.

This leads to a practical framework of **Adaptive Interpretability**. Our model does not force every prediction through a fully interpretable pathway, which could limit performance. Instead, it learns to rely on interpretable, predefined atoms for cases that align with established medical knowledge, while leveraging the expressive power of the ”black-box” neural module for more complex or novel cases. The contribution ratios shown in Figure 3 are a direct measurement of this adaptive behavior. This hybrid approach offers a pragmatic solution to the performance-versus-interpretability dilemma in clinical AI, where the goal is not just to be right, but to be right for the right, understandable reasons, especially in high-stakes decisions.

VI. LIMITATIONS AND FUTURE WORK

While our study provides foundational insights into DiffFNN’s application in medicine, we acknowledge several limitations that open avenues for future work.

1. Interpretability Validation: A primary limitation is that our interpretability analysis remains computational. While we demonstrate that the model can generate traceable rules and adaptively uses its interpretable atoms, the practical clinical utility of these explanations has not been formally validated. As both reviewers rightly noted, a formal user study with clinicians is a critical and necessary next step to empirically verify whether the generated explanations are genuinely meaningful, trustworthy, and useful in a clinical setting.

2. Limited Baseline Coverage: The current study does not include comparisons against the latest state-of-the-art architectures, such as Transformer-based models or more sophisticated GNNs. This was a deliberate choice to focus our analysis on the unique properties of the fuzzy-neural hybrid paradigm. However, we acknowledge that such comparisons are essential for contextualizing our model’s performance against the absolute state of the art. Future work should include these advanced baselines to provide a more complete performance landscape.

3. Dataset Scale and Generalizability: Our evaluation used established, relatively small, and well-structured datasets, necessary for reliable results and rigorous validation. However, real-world clinical data is often larger, sparser, and noisier. Extending our evaluation to large-scale clinical datasets (e.g., from electronic health records) is crucial to test scalability and robustness.

4. Hyperparameter Optimization and Reproducibility: Due to computational constraints, we used fixed hyperparameters based on common practices. Systematic, automated optimization could further enhance performance. To address reproducibility concerns, we commit to making the full source code, preprocessing scripts, and hyperparameter configurations publicly available upon publication.

5. Advanced Fusion Methods: Our implementation uses simple concatenation to combine predefined and learned atoms for transparency. Exploring more sophisticated fusion methods, such as gating mechanisms or cross-attention, could potentially improve performance while maintaining a degree of interpretability and is a promising direction for future research.

VII. Conclusion

This paper successfully extended and evaluated Differentiable Fuzzy Neural Networks [3] for medical drug recommendation, analyzing its performance and interpretability across three distinct medical tasks. Our empirical validation reveals task-dependent performance patterns: DiffFNN shows significant performance gain on adverse reaction prediction, and achieves competitive results on drug repurposing and therapeutic target prediction, providing a highly interpretable framework without compromising performance.

The neural enhancement module addresses DiffFNNś limitation of fixed features by incorporating learnable components while preserving interpretability through selective weighting. This work establishes that the model effectively balances interpretable atoms with learned features—a critical requirement for clinical applications.

For medical AI practitioners, this study provides foundational evidence that hybrid fuzzy-neural approaches can achieve both competitive performance and clinical interpretability in controlled settings. The success of our model depends on aligning model design with task characteristics: fuzzy logic for uncertainty gradations, neural enhancement for complex relationships. Having established the methodś effectiveness on well-defined therapeutic categories, future work should extend this approach to full-scale medical datasets, moving from initial validation toward production-ready systems.

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